

Charan Vardham

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EDUCATION

Case Western Reserve University

Bachelor of Science in Computer Science, Minor in Math

Cleveland, OH

Aug. 2024 – May 2028

Relevant Courses

Intro to Data Structures, Discrete Mathematics, Programming in Java, Algorithms, Computer Security, Linear Algebra

EXPERIENCE

AI/ML Intern

May 2025 - Aug 2025

IQVIA

Remote

- Built a scalable RAG-based medical chatbot to minimize hallucinations and improve factuality processing over 200MB of clinical documents
- Implemented document ingestion, text chunking, embedding generation, and retrieval pipelines achieving 90% retrieval accuracy with MongoDB-backed chat history.
- Worked on a separate voice-enabled chatbot project, integrating NVIDIA Riva APIs for real-time STT/TTS with an LLM backend

xLab Intern

Jan 2026 – Present

Weatherhead School of Management, CWRU

Cleveland, OH

- Developing an AI-powered case study platform using LLMs to simulate real-world business decision-making.
- Building a full-stack application with Next.js (TypeScript), Tailwind CSS, PostgreSQL, and Prisma.
- Integrating OpenAI/Anthropic APIs and HeyGen avatars for dynamic, interactive user experiences.

Undergraduate Student Researcher

Aug 2025 – Present

Case Western Reserve University

Cleveland, OH

- Investigated deep learning-based kernel transfer methods for CT imaging, synthesizing scans across reconstruction kernels without access to raw CT scan data
- Implemented and trained CNN models on a dataset of 5000+ CT scan slices in PyTorch, evaluating performance using image similarity and reconstruction metrics such as PSNR.
- Ongoing work toward preparing a research paper.

Freshman Outreach Representative

Aug 2024 - Present

Case Western Reserve University

Cleveland, OH

- Designed and implemented a Node.js nutrition app, providing meal recommendations based on dietary intake and fitness goals.
- Led outreach initiatives for the largest computer science organization on campus, engaging 100+ incoming students through weekly workshops, technical sessions, and social events

PROJECTS

Shakespeare GPT | PyTorch, Python, CUDA

January 2026

- Implemented a decoder-only Transformer model from scratch to generate text in the style of Shakespeare
- Implemented Multi-Head Self-Attention, Feed-Forward Networks, and Residual Blocks with Layer Normalization
- Optimized training loop using Cross-Entropy loss and AdamW optimizer, achieving a validation loss of 1.5

Custom BPE Tokenizer | Python, Unicode

January 2026

- Developed a Byte Pair Encoding (BPE) tokenizer mirroring GPT-2's architecture to generate tokens from raw text
- Implemented iterative statistical pair merging to construct a custom vocabulary directly from raw Unicode bytes
- Built encoding and decoding pipelines to convert text into token sequences and vice versa

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, SQL, R, HTML/CSS

Frameworks / Libraries: React, Node.js, Next.js, FastAPI, Material-UI, Pandas, NumPy, Matplotlib, Scikit-Learn, PyTorch, TensorFlow, PyTorch Lightning, Hugging Face Transformers

Developer Tools / Deployment: Git, Docker, VS Code, IntelliJ, PyCharm, Visual Studio, REST APIs, AWS, CI/CD, Jenkins, Kubernetes

Testing / ML Skills: Unit testing, Data preprocessing, Feature engineering, Model evaluation metrics (PSNR, SSIM), RAG pipelines, LLM integration